Assignment 1: Kinematics

ROSE5710 **Group 11**

Sun Hejia (1155262875) Long Yifan (1155262899)

Liang Zhuolun (1155262896) Wang Kaiyuan (1155269457)

1 D-H parameters

Modified D-H convention; lengths in m and angles in rad. $r = \sqrt{0.019^2 + 0.035^2} = 0.0398246155$, $\beta = \text{atan2}(19, 35) = 0.4973427163$.

1.1 Left leg

Joint order: hip yaw, hip roll, hip pitch, knee, ankle.

${}^{B}\mathbf{T}_{0,L} = \mathbf{R}_z(\pi/2 + \beta), \quad \mathbf{s}_L = [1, -1, -1, -1, -1]^{\mathsf{T}}$.

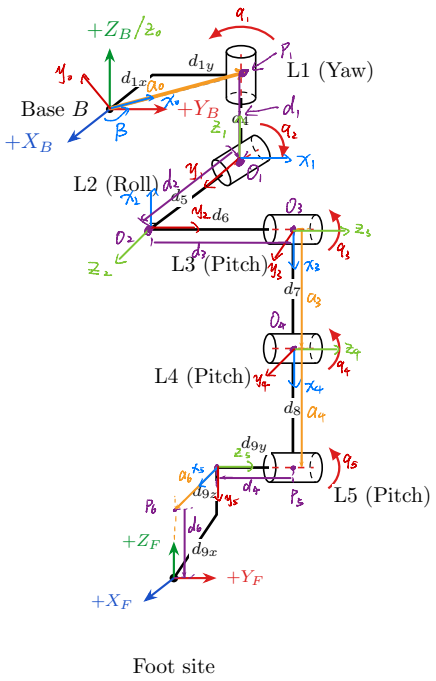


Figure 1: Left leg: D-H origins O_i and intermediate points P_i.

Table 1: Left leg: D–H parameters (m, rad) and zero-pose axes in frame B.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	r	-0.0000741	$q_1 - \beta$
2	$\pi/2$	0	-0.01606	$-q_2 + \pi/2$
3	$\pi/2$	0	0.07415	$-q_3 + \pi$
4	0	0.07865	0	$-q_4$
5	0	0.07865	-0.01975	$-q_5 - \pi/2$
6*	$\pi/2$	0.0005	-0.036225	0

* Fixed foot site.

i	$+x_i$	$+y_i$	$+z_i$
0	$-sX + cY$	$-cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3	$-Z$	$-X$	$+Y$
4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6*	$+X$	$+Y$	$+Z$

X, Y, Z are base-frame unit axes. $s = \sin\beta$, $c = \cos\beta$.

1.2 Right leg

Joint order: hip yaw, hip roll, hip pitch, knee, ankle.

$$ {}^{B}\mathbf{T}_{0,R} = \mathbf{R}_z(-\pi/2 - \beta), \qquad \mathbf{s}_R = [1, -1, 1, -1, -1]^{\mathsf{T}}. $$

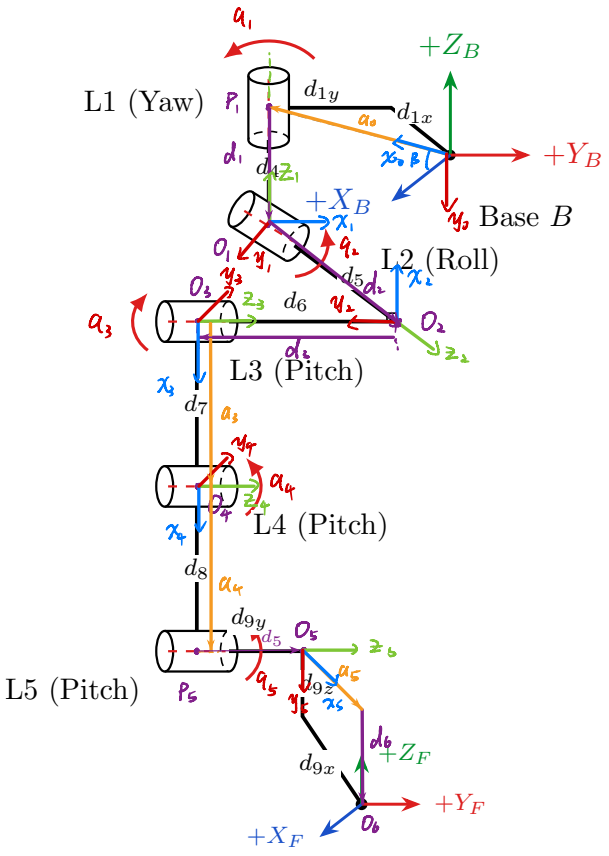


Figure 2: Right leg: D-H origins O_i and intermediate points P_i.

Table 2: Right leg: D–H parameters (m, rad) and zero-pose axes in frame B.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	r	-0.0000741	$q_1 + \pi + \beta$
2	$\pi/2$	0	-0.01606	$-q_2 + \pi/2$
3	$\pi/2$	0	-0.07415	$q_3 + \pi$
4	0	0.07865	0	$-q_4$
5	0	0.07865	0.01735	$-q_5 - \pi/2$
6*	$\pi/2$	0.0005	-0.036225	0

* Fixed foot site.

i	$+x_i$	$+y_i$	$+z_i$
0	$-sX - cY$	$cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3	$-Z$	$-X$	$+Y$
4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6*	$+X$	$+Y$	$+Z$

X, Y, Z are base-frame unit axes. $s = \sin\beta$, $c = \cos\beta$.

1.3 Head and neck

Joint order: neck pitch, head pitch, head yaw, head roll.

$$ {}^{B}\mathbf{T}_{0,H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{p}_c \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}, \qquad \mathbf{p}_c = \begin{bmatrix} 0.001 \\ 0.01915 \\ 0.0900009 \end{bmatrix} \text{m}, \qquad \mathbf{s}_H = [1,1,1,1]^\mathsf{T}. $$

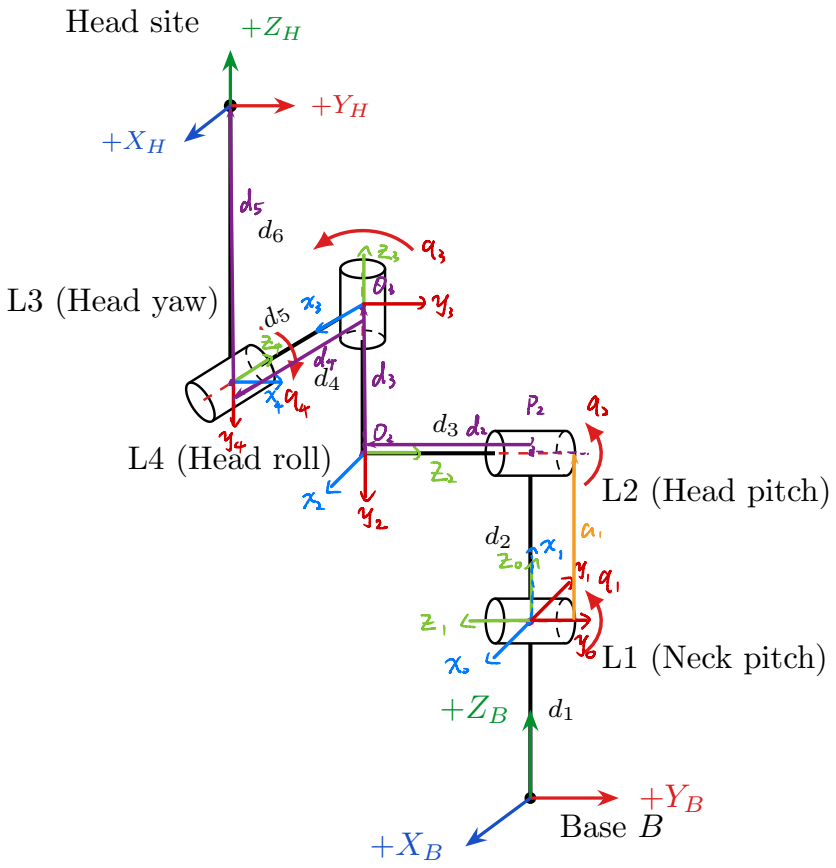


Figure 3: Head: D-H origins O_i and intermediate points P_i.

Table 3: Head: D–H parameters (m, rad) and zero-pose axes in frame B.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	$\pi/2$	0	0	$q_1 + \pi/2$
2	0	0.066	0.01905	$q_2 - \pi/2$
3	$-\pi/2$	0	0.03785	$q_3 + \pi/2$
4	$-\pi/2$	0	-0.005	q_4
5*	$\pi/2$	0	0.04245	$-\pi/2$

* Fixed head site.

i	$+x_i$	$+y_i$	$+z_i$
0	$+X$	$+Y$	$+Z$
1	$+Z$	$-X$	$-Y$
2	$+X$	$+Z$	$-Y$
3	$+Y$	$-X$	$+Z$
4	$+Y$	$-Z$	$-X$
5*	$+X$	$+Y$	$+Z$

X, Y, Z are base-frame unit axes. $O_0 = O_1$, but their axis directions differ.

2 MuJoCo comparison

FK. Each chain uses 100 random configurations within joint limits (seed 0). Limits: 10^{-6} m position and 10^{-5} rad orientation.

Table 4: Measured FK maximum errors over 300 random configurations.

Chain	Max. position (m)	Max. orientation (rad)	Result
Left leg	8.455733×10^{-15}	2.283510×10^{-14}	PASS
Right leg	4.429498×10^{-14}	2.375025×10^{-14}	PASS
Head	6.572043×10^{-15}	1.032631×10^{-13}	PASS

IK. The supplied trajectories use five cycles of 101 phases per chain (1,515 poses). Limits: 10^{-5} m position and 10^{-4} rad orientation.

Table 5: Measured IK maximum errors over 1,515 trajectory poses.

Chain	Max. position (m)	Max. orientation (rad)	Result
Left leg	9.847942×10^{-6}	6.846493×10^{-5}	PASS
Right leg	9.844913×10^{-6}	7.038121×10^{-5}	PASS
Head	8.393151×10^{-6}	6.773237×10^{-5}	PASS

IK settings: central-difference $h = 10^{-6}$, step 0.1, at most 5,000 iterations; stopping at weighted pose cost $E < 10^{-8}$. All three chains pass both FK and IK limits.

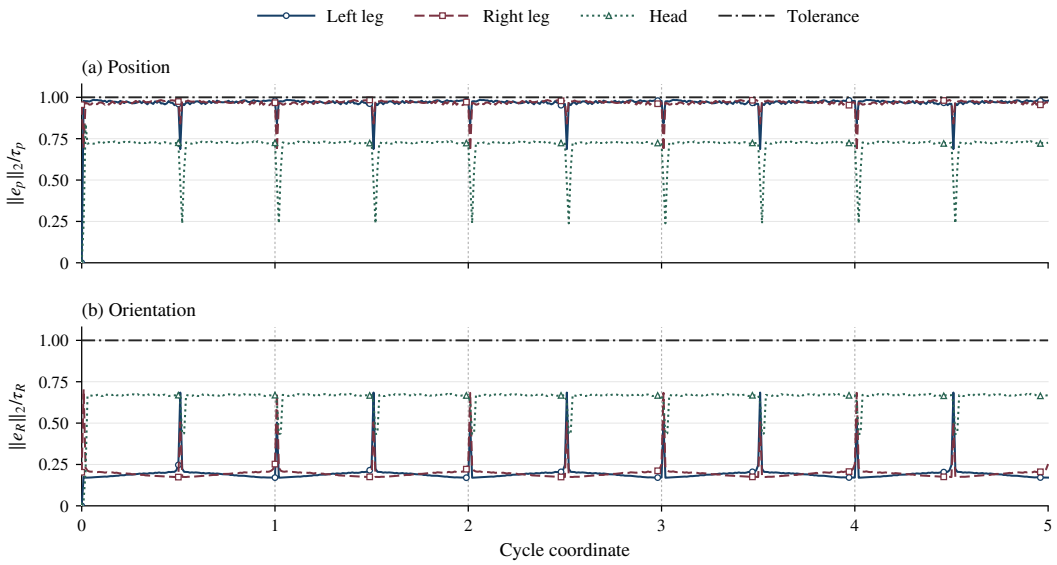


Figure 4: Position and orientation errors over five IK cycles, normalized by their limits. Values below one pass.